Technology Trends Analysis using Job Market & Developer Survey Data

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EXECUTIVE SUMMARY

1. A thorough analysis of technology

This project uses data from developer surveys and job advertisements to analyse worldwide technological trends.

2. Present-Day Market Leaders

The two most popular technologies at the moment are Python and JavaScript.

- In terms of usage overall, JavaScript is the most popular.
- Python is in high demand in many different fields.
- SQL is still a fundamental technological ability.

3. Prospects for Demand

According to survey results, Python is anticipated to drive future technological demand.

4. The popularity of databases

The most popular and extensively used database system is still PostgreSQL.

5. International Involvement & Perspectives

The interactive dashboard offers clear insights into current usage and potential trends, with the United States having the most responders.

INTRODUCTION

Objective:

to use job postings and global developer surveys to assess current and future technological demand.

to determine which databases, tools, and programming languages are most in-demand.

to comprehend the ways in which preferences for technology are changing throughout time.

Target Audience:

Students who want to increase their employment prospects.

Job searchers match market need with their talents.

hiring managers who keep an eye on new technology.

Decision-makers in business plan the adoption of technology and personnel.

Value of the Analysis:

identifies technological skills that are in need right now.

supports hiring and career decisions based on data.

draws attention to upcoming technological developments for competitiveness.

uses interactive dashboards to deliver lucid insights.

METHODOLOGY

Data Sources

- **Job Postings API** → `job-postings.xlsx`
- **Web-scraped salary dataset** → `popular-languages.csv`
- **Stack Overflow Developer Survey** → processed into `survey_dashboard_ready.csv`

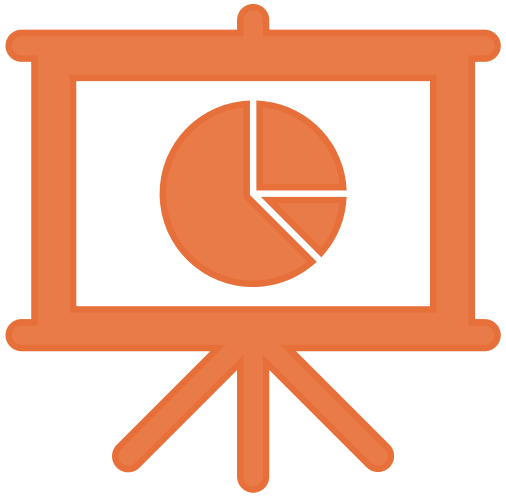
Data Cleaning

- Null values were eliminated.
- Divide columns with multiple values.
- standardized names for technologies
- conducted a frequency analysis.

Tools Used

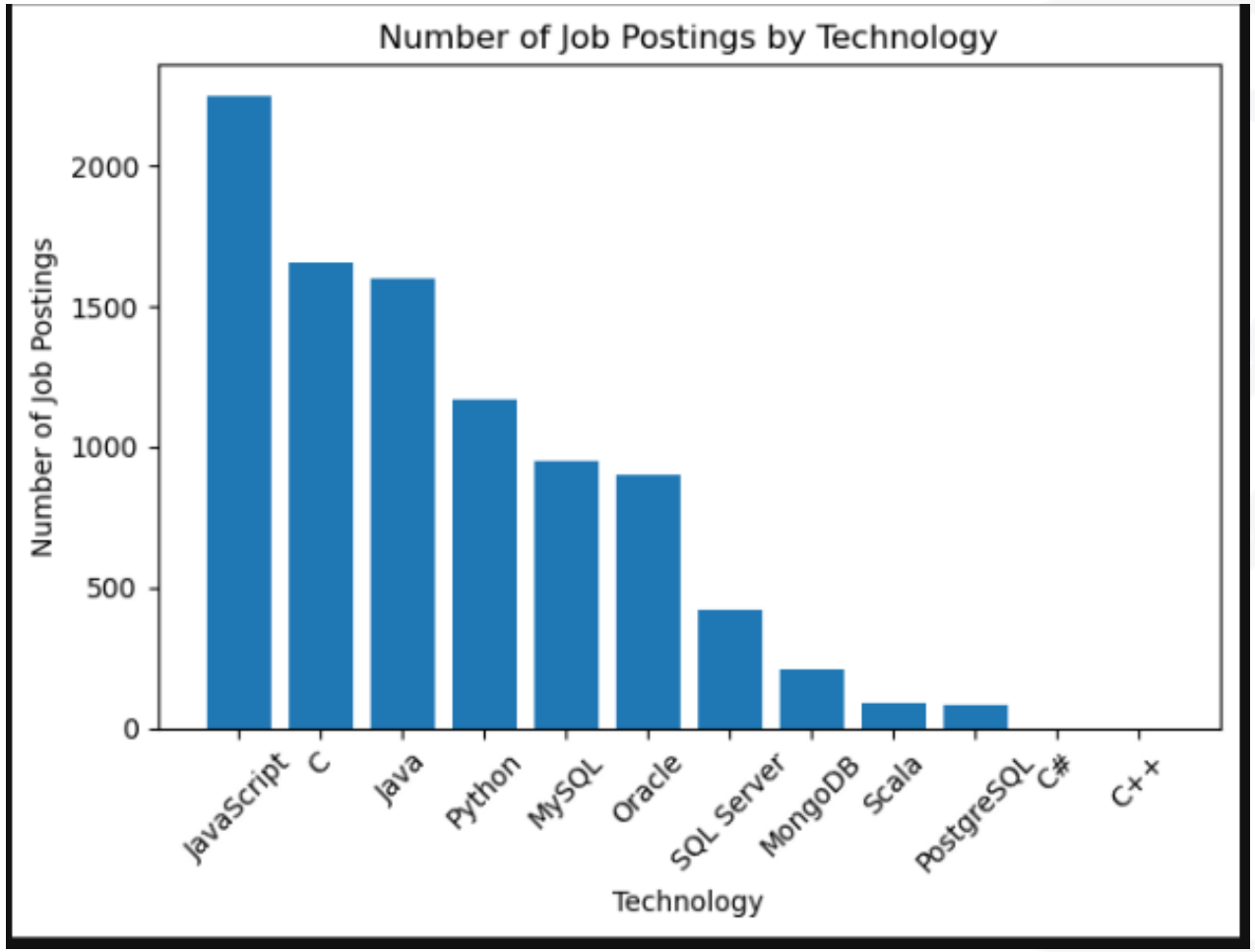
- Data analysis and cleaning using Python (Pandas, Matplotlib)
- IBM Cognos Analytics: Creating dashboards
- Microsoft Excel: Preliminary data analysis and formatting

Results



- The analytical results based on survey data and job ads are shown in the following slides.
- Visualizations draw attention to important developments in databases, tools, and computer languages.
- Dashboard analytics support insights.

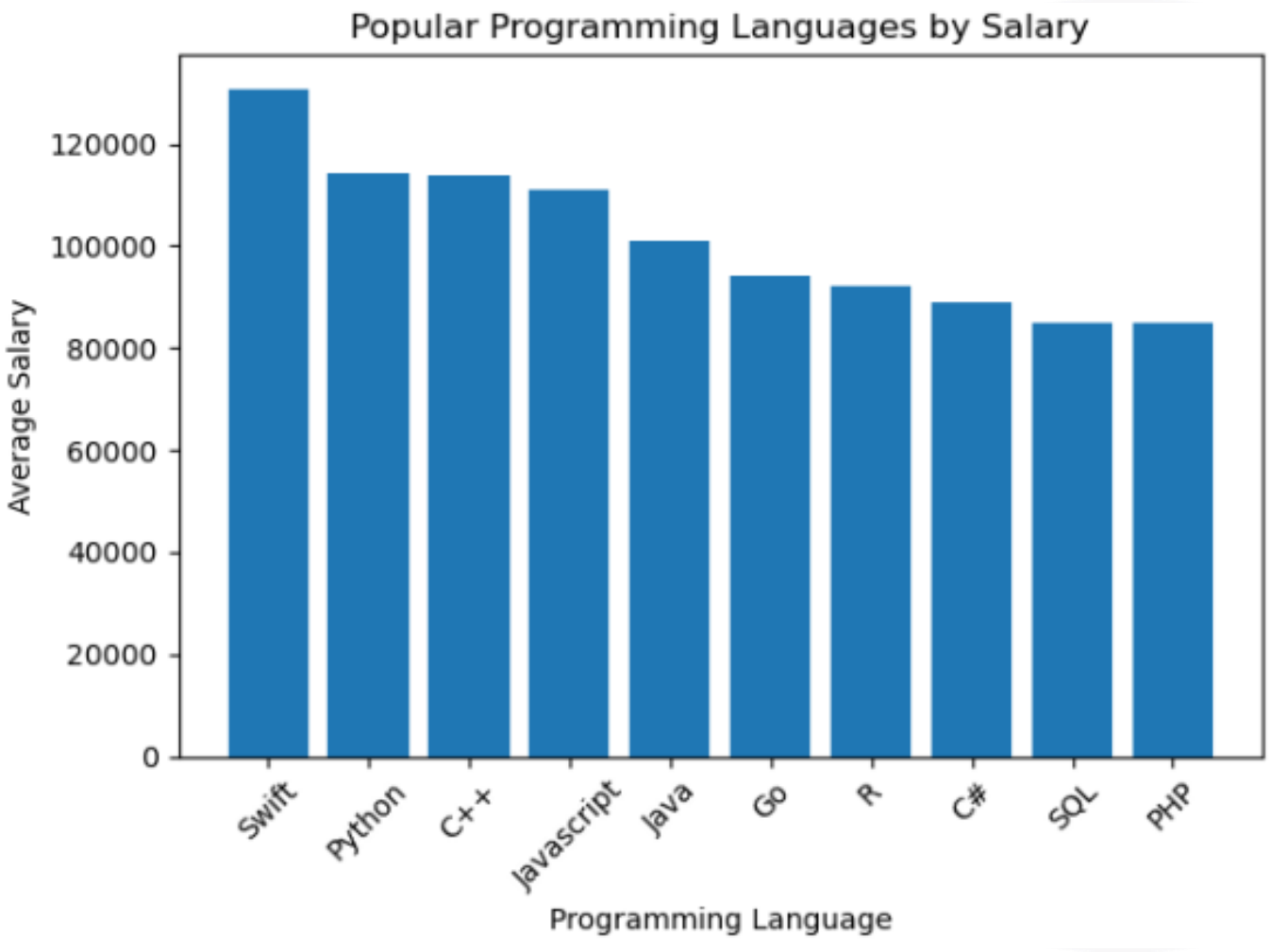
JOB POSTINGS



- JavaScript has the highest number of job postings.
- C and Java also show strong demand.
- SQL and Python remain important skills.
- C++ shows comparatively lower postings.

Insight:
Frontend and full stack skills dominate hiring demand.

POPULAR LANGUAGES

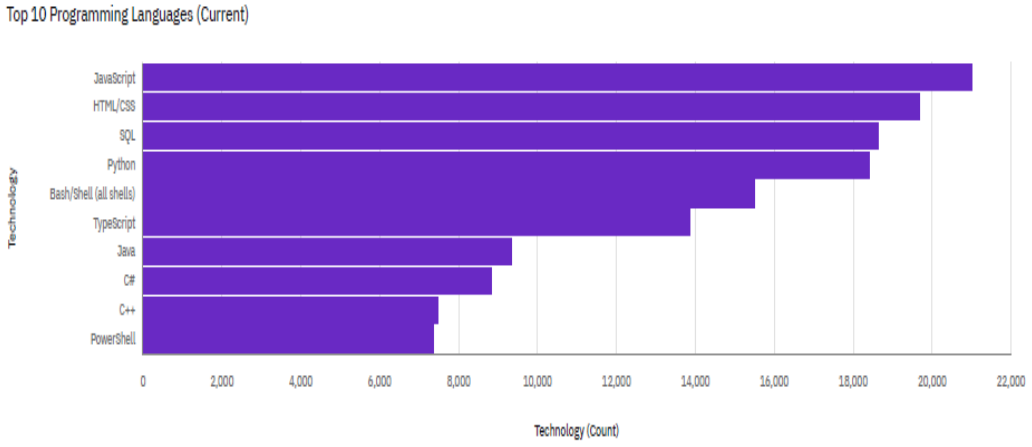


- Swift shows the highest average salary.
- Python and C++ offer strong compensation.
- Java and Go also remain competitive.
- PHP and SQL show relatively lower salary averages.

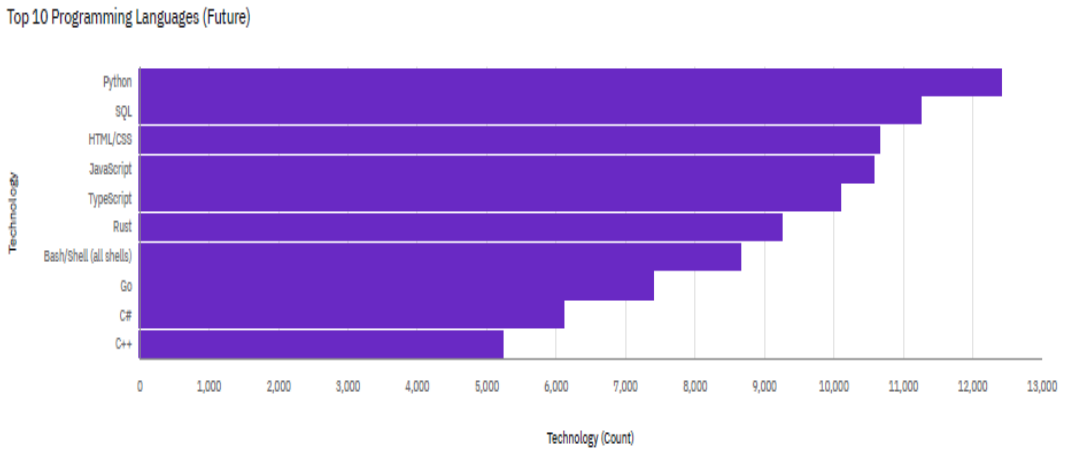
Insight:
High salary does not always mean highest demand.

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



Current:

- JavaScript leads usage.
- HTML/CSS and SQL remain essential.
- Python is among top 4.

Future:

- Python moves to first position.
- SQL remains strong.
- Rust and Go appear in top rankings.

PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

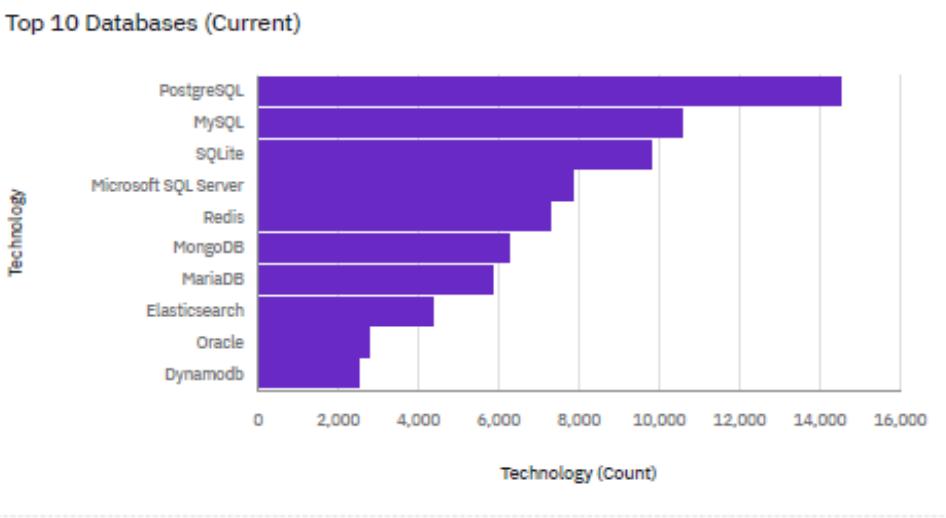
- The need for Python is rising.
- JavaScript is still very important.
- Rust and Go are emerging technologies.

Implications

- Learning Python increases job opportunities.
- Full stack skills remain valuable.
- Developers should monitor emerging technologies.

DATABASE TRENDS

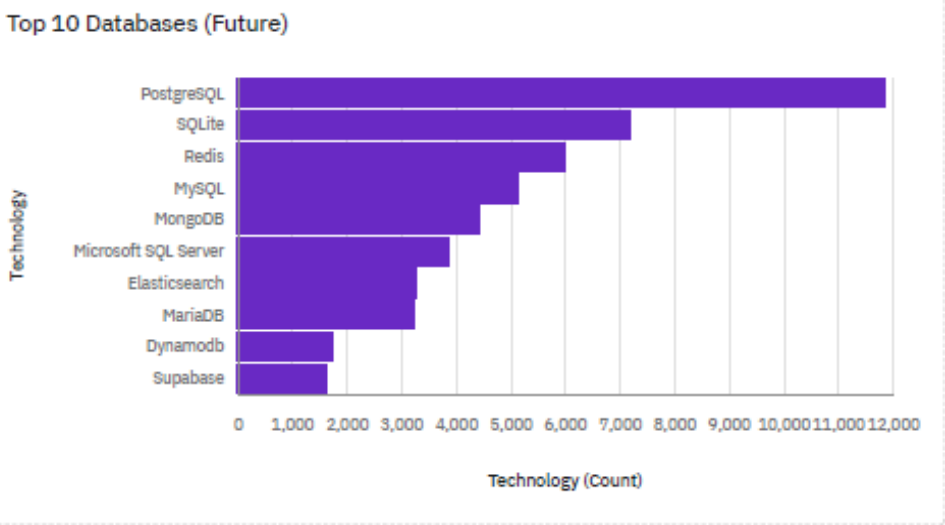
Current Year



Current:

- PostgreSQL leads.
- MySQL and SQLite remain popular.
- SQL Server maintains strong presence.

Next Year



Future:

- PostgreSQL remains dominant.
- Redis and SQLite show growth.
- Supabase appears as emerging database.

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is most preferred database.
- Relational databases still dominate.
- NoSQL databases are growing.

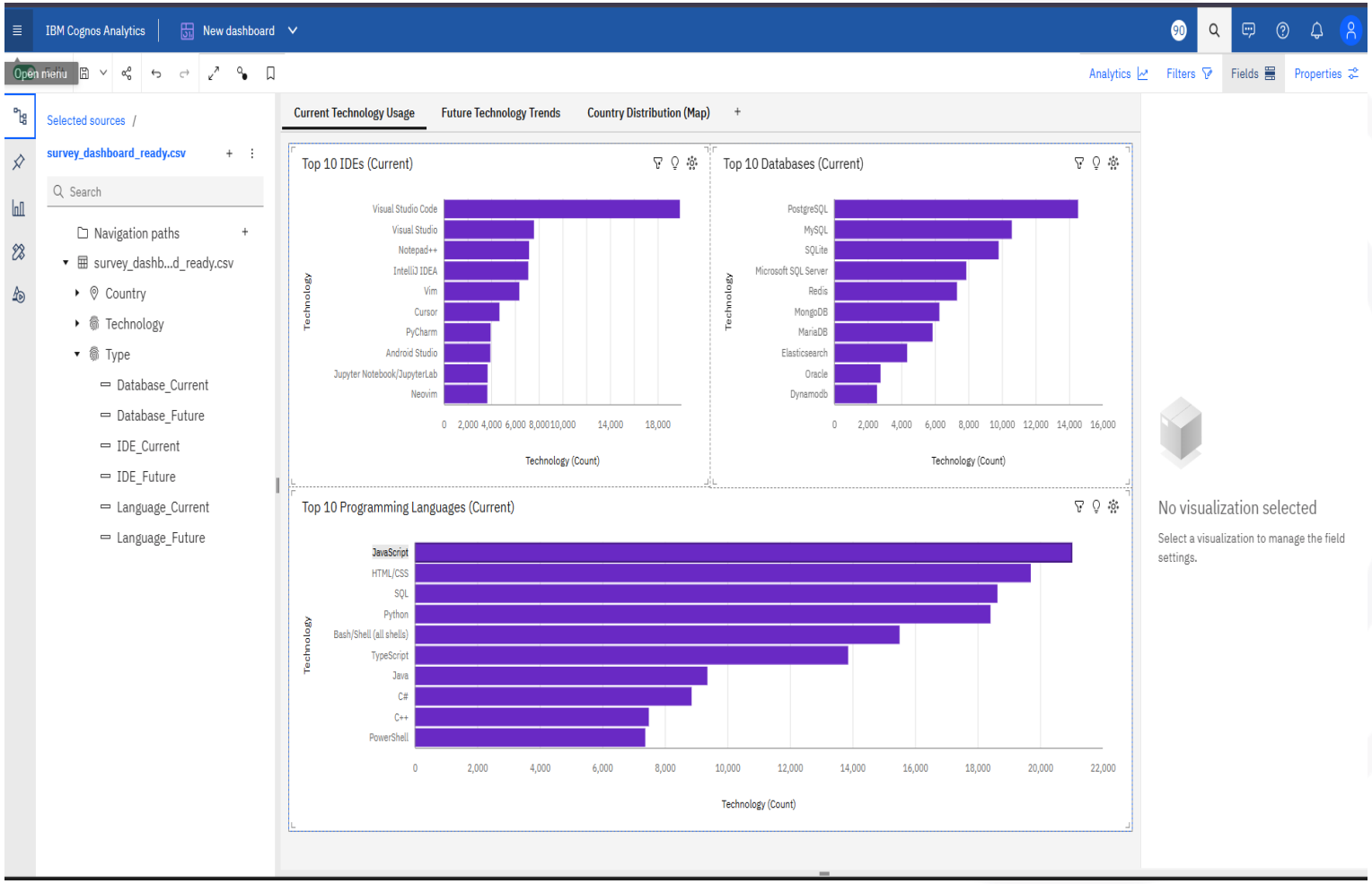
Implications

- SQL skills remain essential.
- Knowledge of both SQL and NoSQL is valuable.
- Cloud native databases are emerging.

DASHBOARD

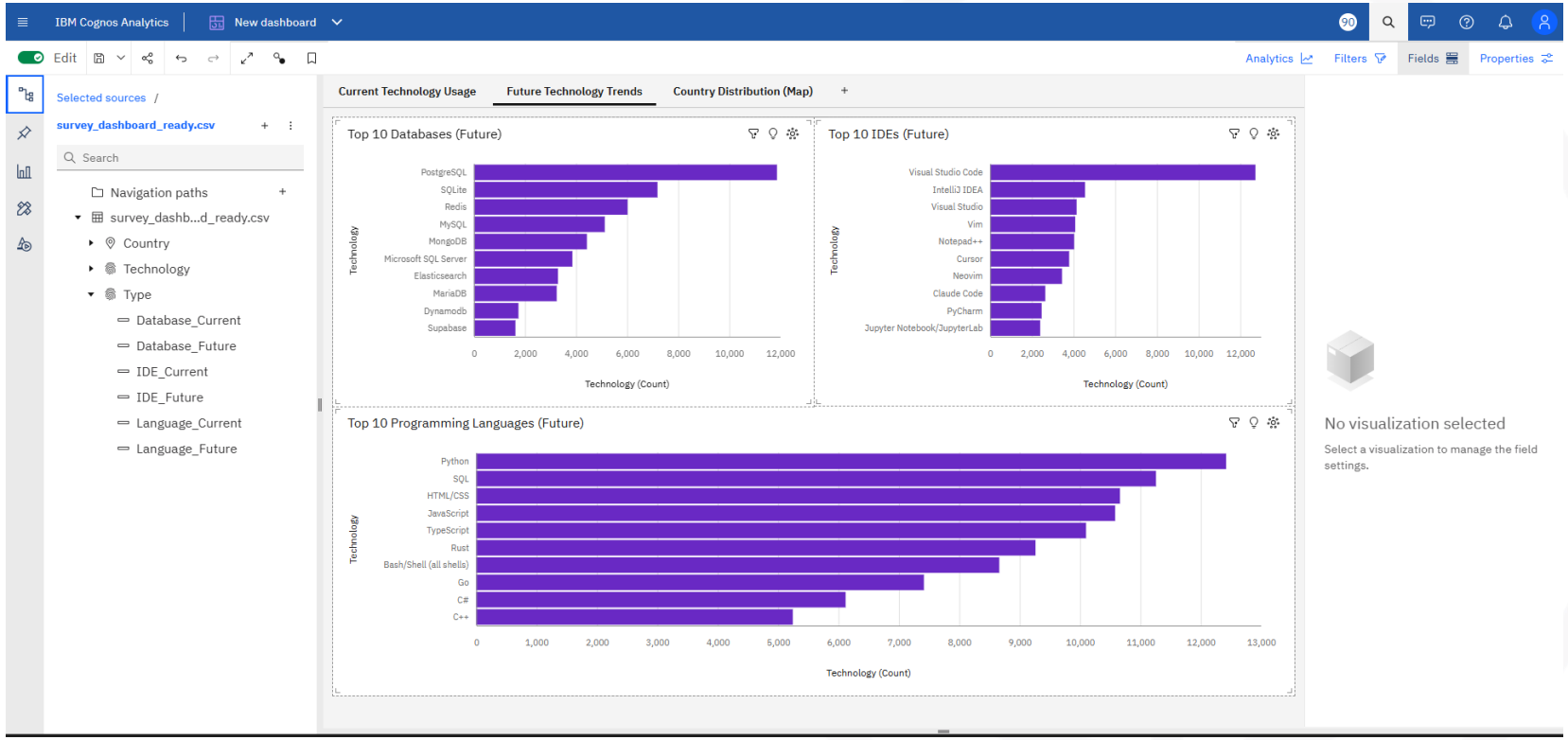
<The next slides present the analytical dashboard developed using IBM Cognos Analytics>

DASHBOARD TAB 1: Current Technology Usage



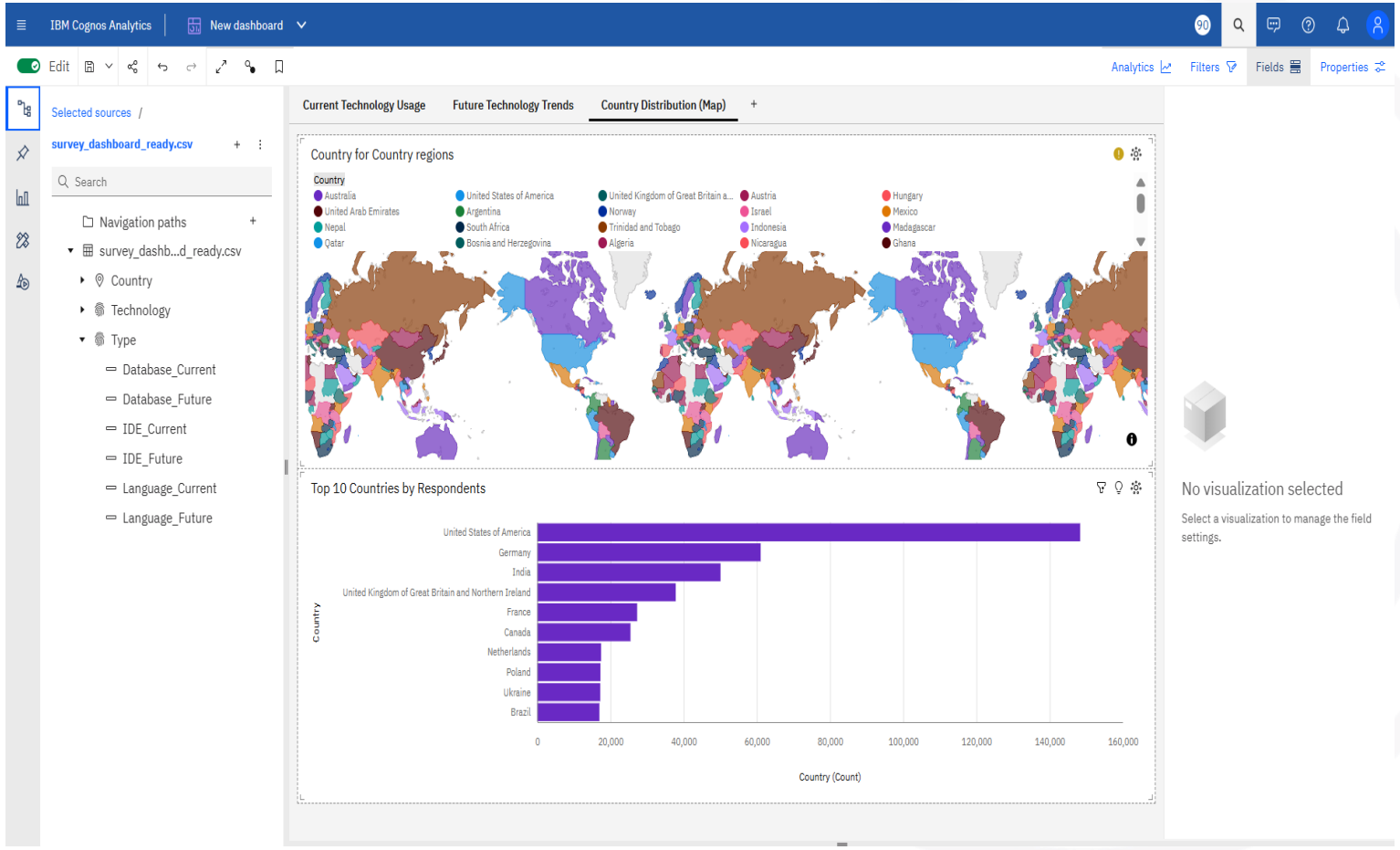
- **JavaScript** is the most widely used programming language among developers.
- **Visual Studio Code** is the most preferred IDE.
- **PostgreSQL** is the leading database technology.
- The **United States** has the highest number of survey respondents.

DASHBOARD TAB 2: Future Technology Trends



- **Python** emerges as the most desired programming language.
- **PostgreSQL** continues to be the top preferred database.
- **Visual Studio Code** remains the leading IDE.
- **Rust** and **Go** show strong growth potential among developers.

DASHBOARD TAB 3: Demographics



- The **United States** leads in overall survey participation.
- **Germany** and **India** follow with significant response counts.
- **Europe** and **North America** dominate overall contributions.
- The data reflects broad global developer trends and technology adoption patterns.

DISCUSSION

Data oriented and backend technologies are showing a clear upward trend based on the analysis. While JavaScript currently leads in overall usage, Python is emerging as the most desired language for the future, indicating a shift toward data driven and backend development roles. PostgreSQL and other open source databases continue to gain strong preference among developers. These regional and technological patterns are clearly visualized and supported through the interactive dashboard insights.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Python is the fastest growing language.
- PostgreSQL dominates database preference.
- JavaScript remains industry standard.

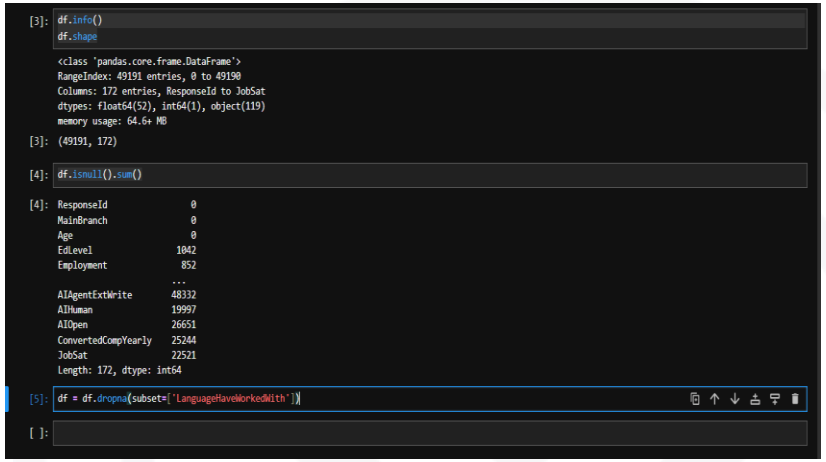
Implications

- Professionals should focus on Python + SQL.
- Organizations should adopt scalable open source databases.
- Continuous learning is essential in tech industry.

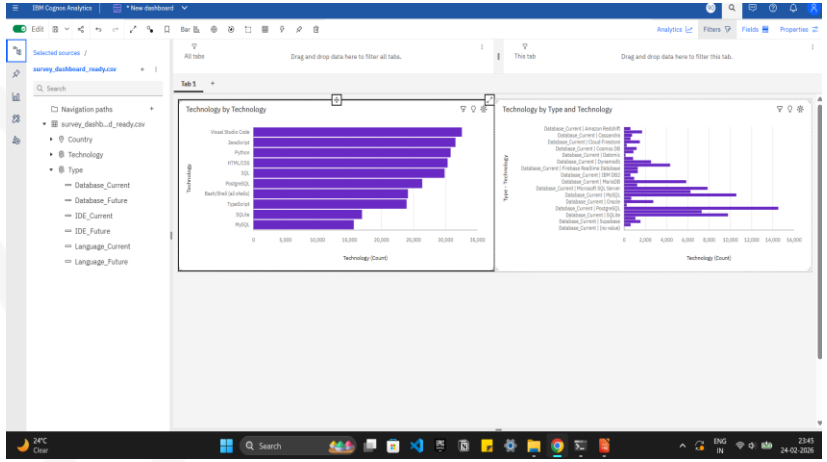
CONCLUSION

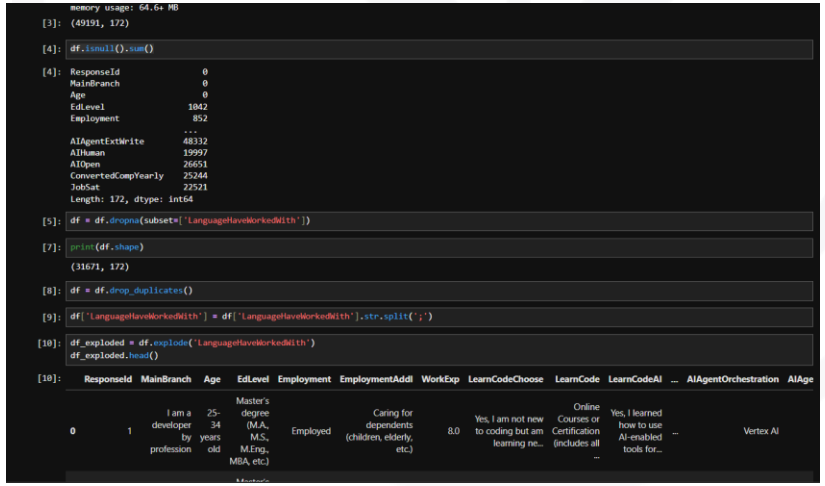
- Data focused and backend technologies are steadily gaining popularity in the tech industry.
- Python is emerging as a leading language for future development trends.
- PostgreSQL continues to be recognized as a reliable and widely preferred database system.
- Dashboard analytics supports informed, data driven decision making through clear visual insights.

APPENDIX



```
[3]: df.info()
     df.shape

     <class 'pandas.core.frame.DataFrame'>
     RangeIndex: 49191 entries, 0 to 49190
     Columns: 172 entries, ResponseId to JobSat
     dtypes: float64(52), int64(1), object(119)
     memory usage: 64.6+ MB

[3]: (49191, 172)

[4]: df.isnull().sum()

[4]: ResponseId              0
     MainBranch              0
     Age                     0
     EdLevel              1042
     Employment            852
                          ...
     AIAgentExtWrite     48332
     AIHuman             19997
     AIOpen              26651
     ConvertedCompYearly 25244
     JobSat              22521
     Length: 172, dtype: int64

[5]: df = df.dropna(subset=['LanguageHaveWorkedWith'])

[ ]:
```





```
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     AIAgentExtWrite     48332
     AIHuman             19997
     AIOpen              26651
     ConvertedCompYearly 25244
     JobSat              22521
     Length: 172, dtype: int64

[5]: df = df.dropna(subset=['LanguageHaveWorkedWith'])

[7]: print(df.shape)

     (31671, 172)

[8]: df = df.drop_duplicates()

[9]: df['LanguageHaveWorkedWith'] = df['LanguageHaveWorkedWith'].str.split(';')

[10]: df_exploded = df.explode('LanguageHaveWorkedWith')
      df_exploded.head()
```

[10]:

	ResponseId	MainBranch	Age	EdLevel	Employment	EmploymentAddl	WorkExp	LearnCodeChoose	LearnCode	LearnCodeAI	...	AIAgentOrchestration	AIAge
0	1	I am a developer by profession	25-34 years old	Master's degree (M.A., M.S., M.Eng., MBA, etc.)	Employed	Caring for dependents (children, elderly, etc.)	8.0	Yes, I am not new to coding but am learning ne...	Online Courses or Certification (includes all ...	Yes, I learned how to use AI-enabled tools for...	...	Vertex AI	